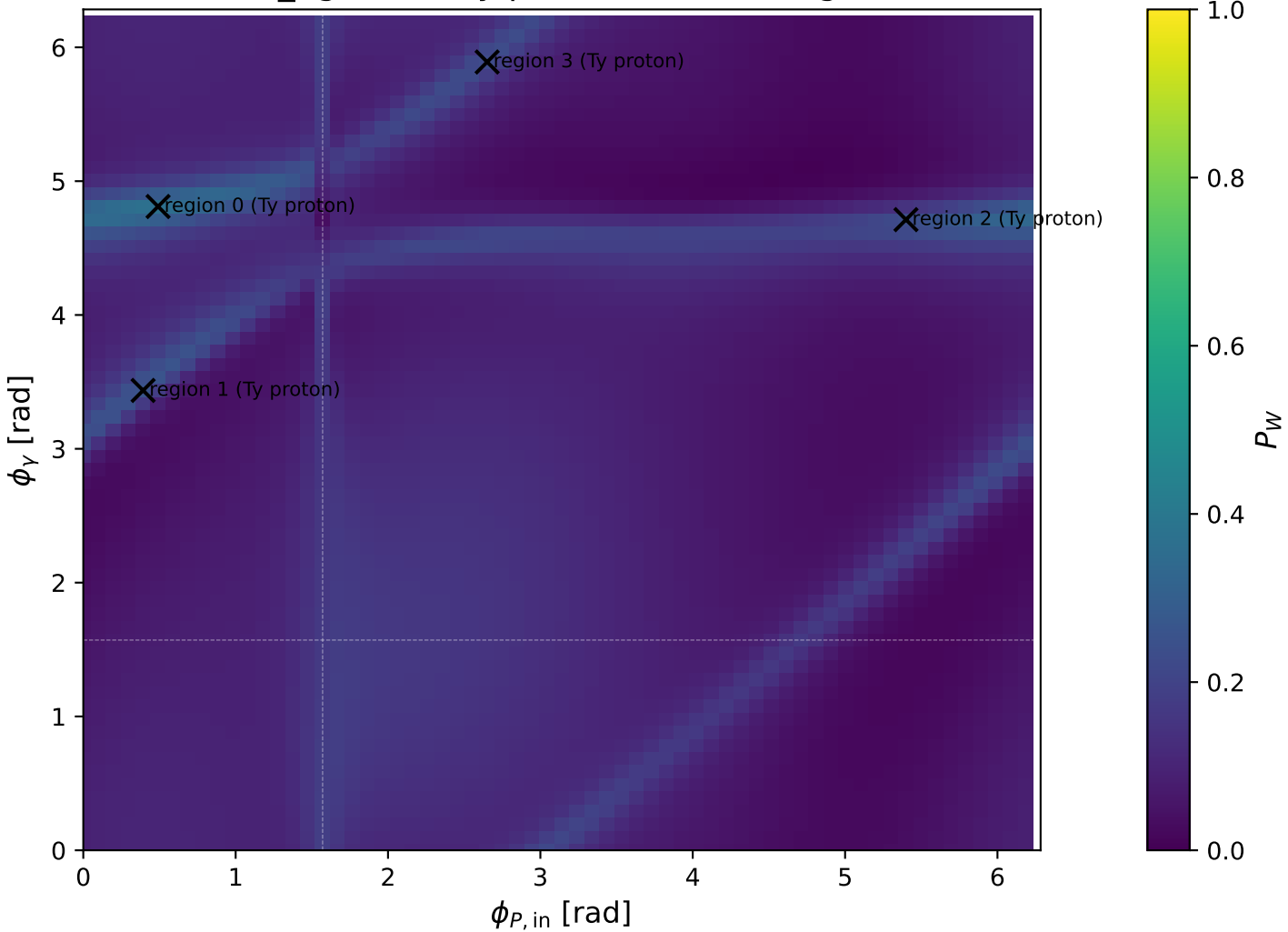


mid_Egamma Ty proton: max P_W regions



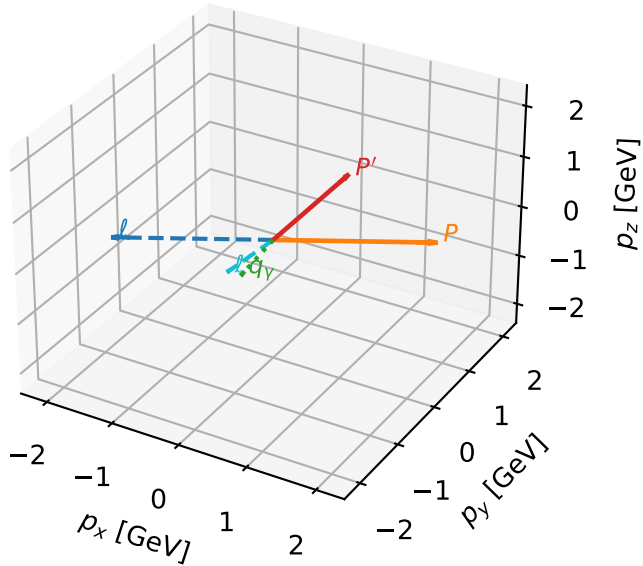
Ty proton characteristicmax P_W configuration

w_purity_mid_egamma_ty_proton_region_0 P_W=0.364153
region: mid_Egamma_Ty_proton_region_0
outgoing-state purity: 0.524494
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_l}\rho(h_l, T_{y,p})$

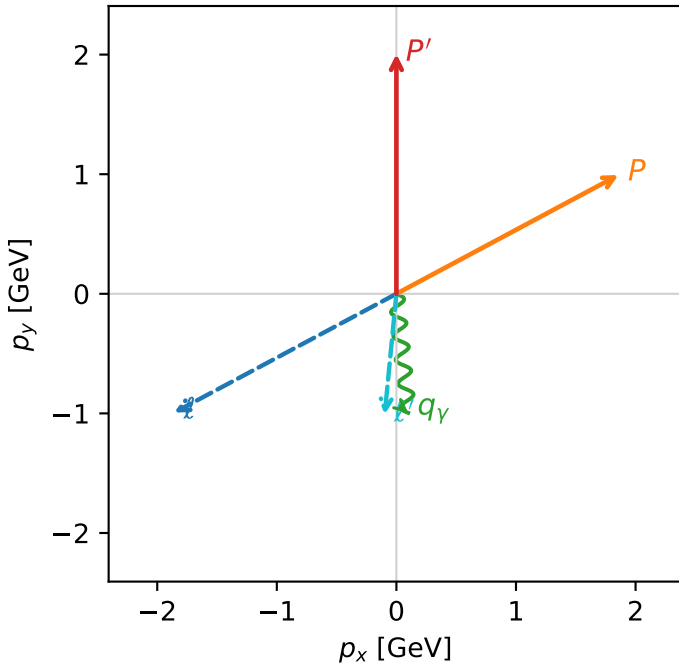
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=2.00525$
 $\theta_{in}=1.57079$, $\phi_l=3.63247$, $\phi_P=0.490874$
 $E_Y=1$, $\phi_Y=4.81056$
 $Q^2=2.27485$, $x_B=0.212729$, $t=-4.46843$, $W^2=9.29862$, $y=0.544593$
 $\theta(\ell', \gamma)=11.1676$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= -1.8582$ $p_y= -0.9932$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 1.8582$ $p_y= 0.9932$ $p_z= 0.0000$
 ℓ' $E= 1.4247$ $p_x= -0.0980$ $p_y= -1.0101$ $p_z= 0.0000$
 P' $E= 2.2138$ $p_x= 0.0000$ $p_y= 2.0053$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= 0.0980$ $p_y= -0.9952$ $p_z= 0.0000$

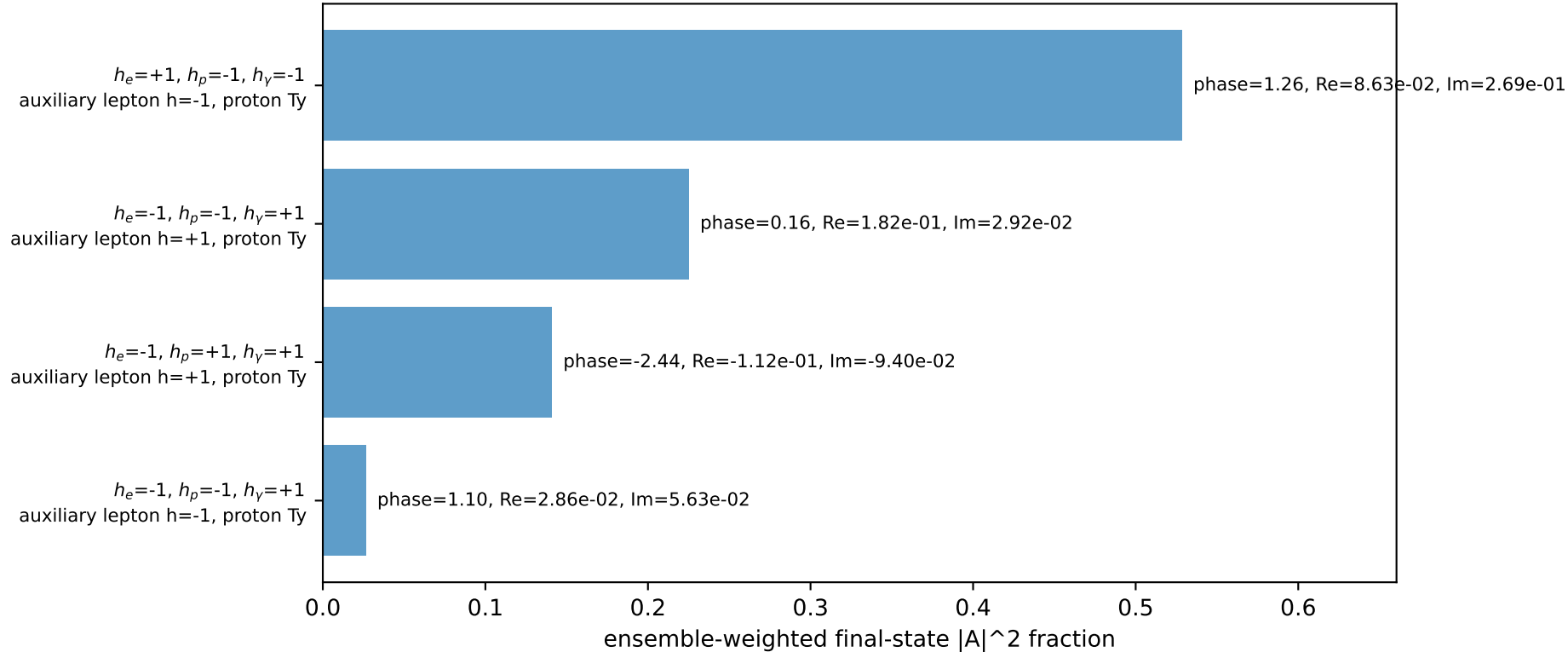
3D momenta



Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=92.1%)

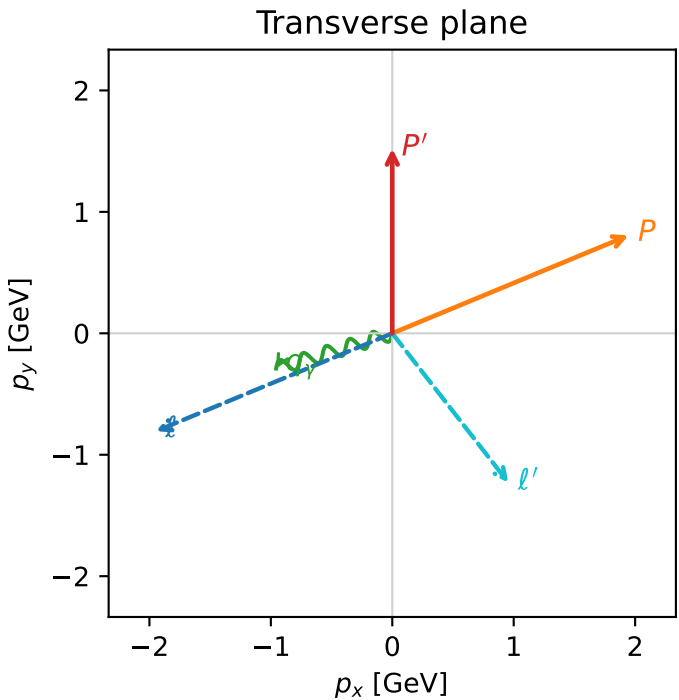
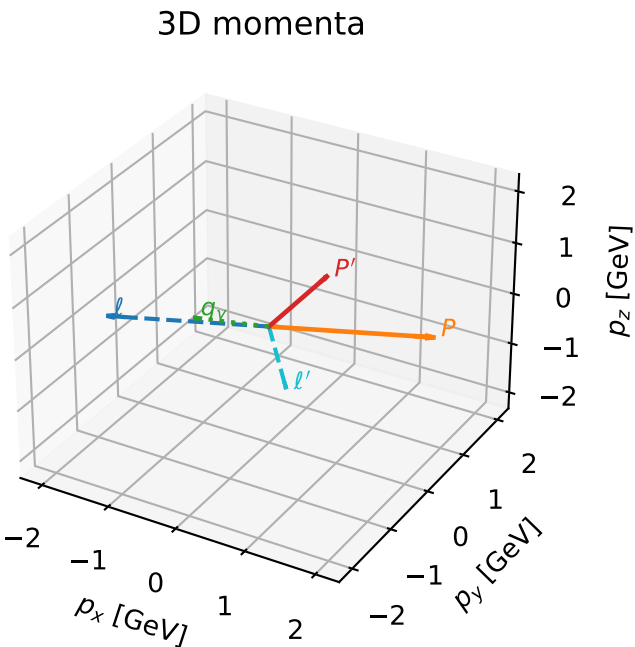


Ty proton characteristicmax P_W configuration

w_purity_mid_egamma_ty_proton_region_1 P_W=0.255535
region: mid_Egamma_Ty_proton_region_1
outgoing-state purity: 0.526591
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_\ell}\rho(h_\ell,T_{y,p})$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.52061$
 $\theta_{in}=1.57079$, $\phi_\ell=3.53429$, $\phi_p=0.392699$
 $E_\gamma=1$, $\phi_\gamma=3.43612$
 $Q^2=8.3794$, $x_B=0.65283$, $t=-4.0293$, $W^2=5.33595$, $y=0.653672$
 $\theta(\ell',\gamma)=111.001$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= -1.9466$ $p_y= -0.8063$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 1.9466$ $p_y= 0.8063$ $p_z= 0.0000$
 ℓ' $E= 1.8519$ $p_x= 0.9569$ $p_y= -1.2303$ $p_z= 0.0000$
 P' $E= 1.7866$ $p_x= 0.0000$ $p_y= 1.5206$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.9569$ $p_y= -0.2903$ $p_z= 0.0000$

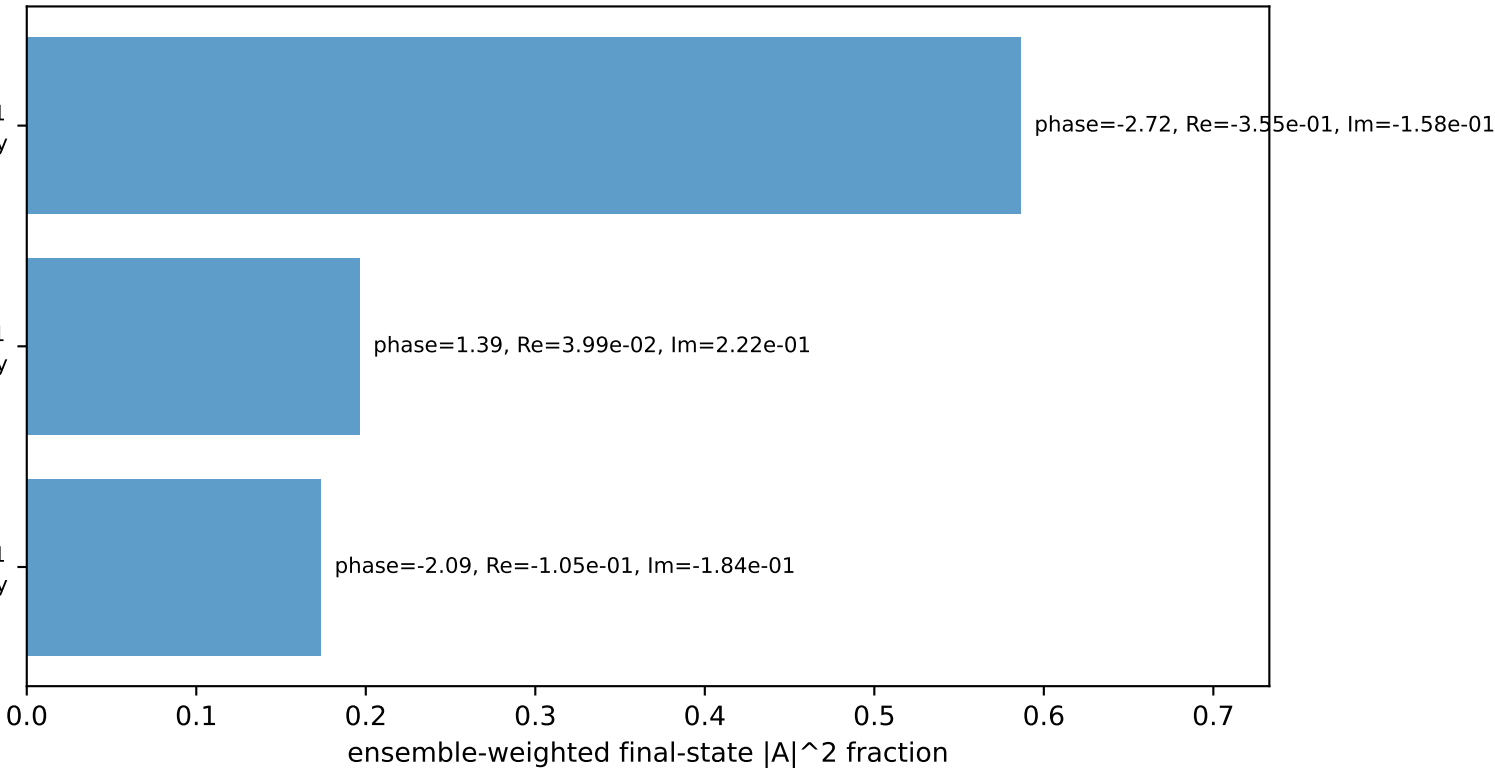


$h_e=-1, h_p=-1, h_\gamma=+1$
auxiliary lepton $h=+1$, proton Ty

$h_e=+1, h_p=+1, h_\gamma=-1$
auxiliary lepton $h=-1$, proton Ty

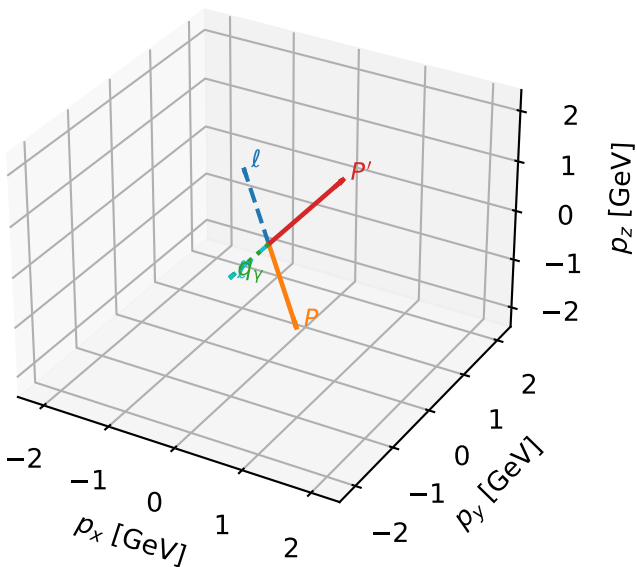
$h_e=+1, h_p=-1, h_\gamma=-1$
auxiliary lepton $h=-1$, proton Ty

Leading initial-ensemble/final-state components (N=3, retained=95.6%)



Ty proton characteristic P_W configuration

3D momenta

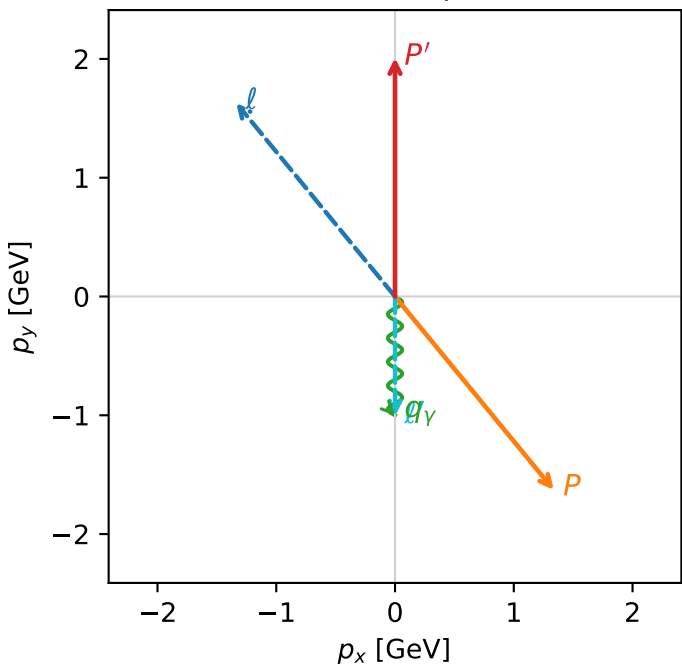


w_purity_mid_egamma_ty_proton_region_2 P_W=0.231192
 region: mid_Egamma_Ty_proton_region_2
 outgoing-state purity: 0.504733
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $\frac{1}{2} \sum_{h_\ell} \rho(h_\ell, T_{Y,p})$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=2.00946$
 $\theta_{in}=1.57079$, $\phi_\ell=2.25802$, $\phi_P=5.39961$
 $E_Y=1$, $\phi_Y=4.71239$
 $Q^2=7.91593$, $x_B=0.483562$, $t=-15.0148$, $W^2=9.33396$, $y=0.833675$
 $\theta(\ell', \gamma)=0$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= -1.3366$ $p_y= 1.6287$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 1.3366$ $p_y= -1.6287$ $p_z= 0.0000$
 ℓ' $E= 1.4209$ $p_x= 0.0000$ $p_y= -1.0095$ $p_z= 0.0000$
 P' $E= 2.2176$ $p_x= 0.0000$ $p_y= 2.0095$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= -0.0000$ $p_y= -1.0000$ $p_z= 0.0000$

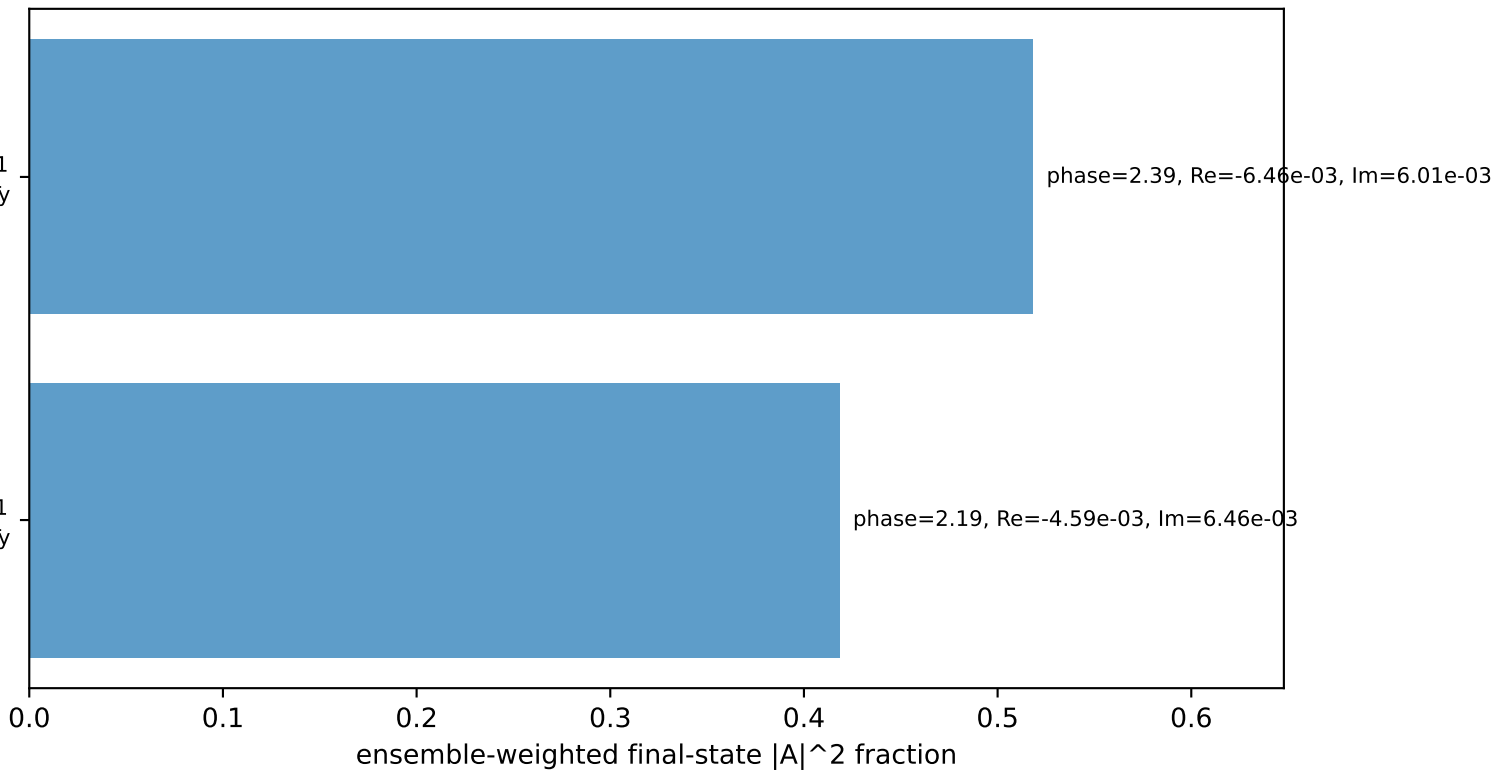
Transverse plane



$h_e=+1, h_p=-1, h_\gamma=-1$
 auxiliary lepton $h=-1$, proton Ty

$h_e=-1, h_p=+1, h_\gamma=+1$
 auxiliary lepton $h=+1$, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=93.7%)

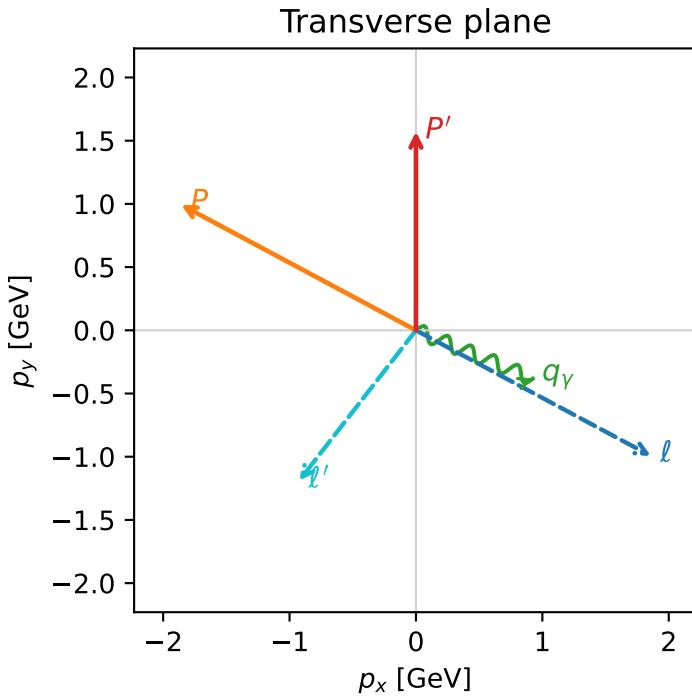
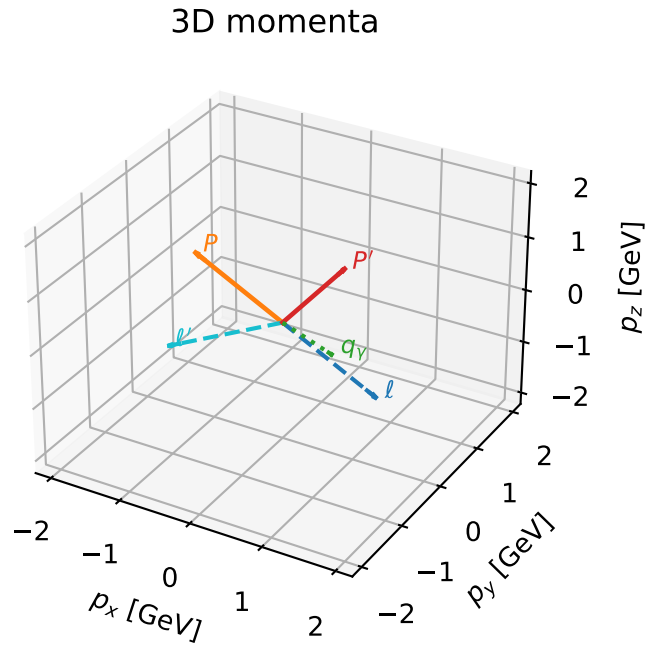


Ty proton characteristic max P_W configuration

w_purity_mid_egamma_ty_proton_region_3 P_W=0.229307
region: mid_Egamma_Ty_proton_region_3
outgoing-state purity: 0.523299
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2} \sum_{h_\ell} \rho(h_\ell, T_{Y,p})$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.57211$
 $\theta_{in}=1.57079$, $\phi_\ell=5.79231$, $\phi_p=2.65072$
 $E_\gamma=1$, $\phi_\gamma=5.89049$
 $Q^2=7.50327$, $x_B=0.606675$, $t=-3.56166$, $W^2=5.74444$, $y=0.629857$
 $\theta(\ell', \gamma)=105.338$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= 1.8582$ $p_y= -0.9932$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -1.8582$ $p_y= 0.9932$ $p_z= 0.0000$
 ℓ' $E= 1.8078$ $p_x= -0.9239$ $p_y= -1.1894$ $p_z= 0.0000$
 P' $E= 1.8307$ $p_x= 0.0000$ $p_y= 1.5721$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.9239$ $p_y= -0.3827$ $p_z= 0.0000$



Leading initial-ensemble/final-state components (N=3, retained=93.7%)

